Name: ____ Period:________________________

AP Chem Unit 4: Equilibrium

Unit Map

Textbook Reference: Zumdahl *Chemistry* Chapter 13

Date	Day	Learning Targets	Activities	Work on this
10/15-16	U4D1	What is equilibrium and how is it related to rates?	• Review for Exam 3 • Penny Ante Lab	Finish U3 Textbook Problem Set #2 Study for test!
10/17-21	U3D8	What have you learned about kinetics?	• U3 Exam	None
10/22-23		PSAT for Juniors	• Wrap up Penny Ante Lab • Review Penny Ante Lab	Finish Penny Ante if needed
10/24-27	U4D2	What is an equilibrium expression?	• Review Exam 3 • Discuss Penny Ante Lab	U4 Textbook Problem Set 1 posted
10/28-29	U4D3	What is equilibrium position?	• Equilibrium expressions, Q vs K, Size of K	Problem set 1 &2?
10/30-31	U4D4	How do systems at equilibrium respond to stresses?	• Equilibrium Practice • Le Chatelier's Principle	Finish Prob Set 1 before quiz next time
11/3-4	U4D5	How do we calculate equilibrium calculations?	• Quiz! • Equilibrium Calculations	Problem Set 2
11/5-6	U4D6	How do we calculate equilibrium calculations?	• More equilibrium calcs	Problem Set 2
11/10-11	U5D1	How do we express energy gained or released in a process?	• Review • Intro to thermochem	Study for test! Prob set #2 due before exam
11/12-13	U4D7	What have you learned about equilibrium?	• Unit 4 Exam	

	Unit 4: General Equilibrium
7.1	
7.1A	**Introduction to Equilibrium**
	Explain the relationship between the occurrence of a reversible chemical or physical process, and the establishment of equilibrium, to experimental observations.
	7.1.A.1: Many observable processes are reversible. Examples include evaporation and condensation of water, absorption and desorption of a gas, or dissolution and precipitation of a salt. Some important reversible chemical processes include the transfer of protons in acid-base reactions and the transfer of electrons in redox reactions.
	7.1.A.2: When equilibrium is reached, no observable changes occur in the system. Reactants and products are simultaneously present, and the concentrations or partial pressures of all species remain constant.
	7.1.A.3: The equilibrium state is dynamic. The forward and reverse processes continue to occur at equal rates, resulting in no net observable change.
	7.1.A.4: Graphs of concentration, partial pressure, or rate of reaction versus time for simple chemical reactions can be used to understand the establishment of chemical equilibrium.
7.2	
7.2A	**Direction of Reversible Reactions**
	Explain the relationship between the direction in which a reversible reaction proceeds and the relative rates of the forward and reverse reactions.
	7.2.A.1: If the rate of the forward reaction is greater than the reverse reaction, then there is a net conversion of reactants to products. If the rate of the reverse reaction is greater than that of the forward reaction, then there is a net conversion of products to reactants. An equilibrium state is reached when these rates are equal.
7.3	
7.3A	**Reaction Quotient and Equilibrium Constant**
	Represent the reaction quotient Qc or Qp, for a reversible reaction, and the corresponding equilibrium expressions Kc = Qc or Kp = Qp.
	7.3.A.1: The reaction quotient Qc describes the relative concentrations of reaction species at any time. For gas phase reactions, the reaction quotient may instead be written in terms of pressures as Qp. The reaction quotient tends toward the equilibrium constant such that at equilibrium Kc = Qc and Kp = Qp. As examples, for the reaction aA + bB ⇌ cC + dD the law of mass action indicates that the equilibrium expression for (Kc, Qc) is **EQN: Kc = [C]c [D]d / [A]a [B]b** and that for (Kp, Qp) is **EQN: Kp = (PC)c (PD)d / (PA)a (PB)b ★ ★**
	7.3.A.2: The reaction quotient does not include substances whose concentrations (or partial pressures) are independent of the amount, such as for solids and pure liquids.
7.4	
7.4A	**Calculating the Equilibrium Constant**
	Calculate Kc or Kp based on experimental observations of concentrations or pressures at equilibrium.
	7.4.A.1: Equilibrium constants can be determined from experimental measurements of the concentrations or partial pressures of the reactants and products at equilibrium.
7.5	
7.5A	**Magnitude of the Equilibrium Constant**
	Explain the relationship between very large or very small values of K and the relative concentrations of chemical species at equilibrium.
	7.5.A.1: Some equilibrium reactions have very large K values and proceed essentially to completion. Others have very small K values and barely proceed at all.
7.6	
7.6A	**Properties of the Equilibrium Constant**
	Represent a multistep process with an overall equilibrium expression, using the constituent K expressions for each individual reaction.
	7.6.A.1: When a reaction is reversed, K is inverted.
	7.6.A.2: When the stoichiometric coefficients of a reaction are multiplied by a factor c, K is raised to the power c.

	7.6.A.3: When reactions are added together, the K of the resulting overall reaction is the product of the K's for the reactions that were summed.
	7.6.A.4: Since the expressions for K and Q have identical mathematical forms, all valid algebraic manipulations of K also apply to Q.
7.7	
7.7A	**Calculating Equilibrium Concentrations**
	Identify the concentrations or partial pressures of chemical species at equilibrium based on the initial conditions and the equilibrium constant.
	7.7.A.1: The concentrations or partial pressures of species at equilibrium can be predicted given the balanced reaction, initial concentrations, and the appropriate K.
	7.7.A.2: When $Q < K$, the reaction will proceed with a net consumption of reactants and generation of products. When $Q > K$, the reaction will proceed with a net consumption of products and generation of reactants. When $Q = K$, the system is at dynamic equilibrium; both forward and reverse reactions proceed at the same rate, and the proportion of reactants and products remains constant.
7.8	
7.8A	**Representations of Equilibrium**
	Represent a system undergoing a reversible reaction with a particulate model.
	7.8.A.1: Particulate representations can be used to describe the relative numbers of reactant and product particles present prior to and at equilibrium, and the value of the equilibrium constant.
7.9	
7.9A	**Introduction to Le Châtelier's Principle**
	Identify the response of a system at equilibrium to an external stress, using Le Châtelier's principle.
	7.9.A.1: Le Châtelier's principle can be used to predict the response of a system to stresses such as addition or removal of a chemical species, change in temperature, change in volume / pressure of a gas-phase system, or dilution of a reaction system.
	7.9.A.2: Le Châtelier's principle can be used to predict the effect that a stress will have on experimentally measurable properties such as pH, temperature, and color of a solution.
7.10	
7.1	**Reaction Quotient and Le Châtelier's Principle**
	Explain the relationships between Q, K, and the direction in which a reversible reaction will proceed to reach equilibrium.
	7.10.A.1: A disturbance to a system at equilibrium causes Q to differ from K, thereby taking the system out of equilibrium. The system responds by bringing Q back into agreement with K, thereby establishing a new equilibrium state.
	7.10.A.2: Some stresses, such as changes in concentration, cause a change in Q only. A change in temperature causes a change in K. In either case, the concentrations or partial pressures of species redistribute to bring Q and K back into equality.

AP Chem Unit 7 Equilibrium Notes

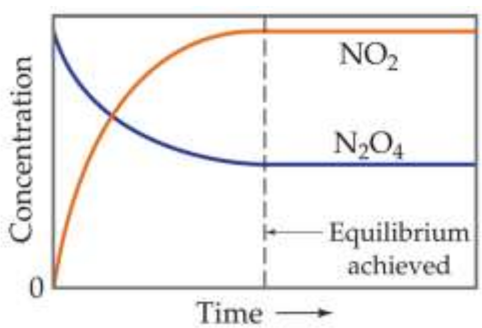
7.1 Intro to Equilibrium
7.2 Direction of Reversible Rxns
7.3 Rxn Quotient and the EQ Constant
7.4 Calculating the Equilibrium Constant
7.6 Properties of the Equilibrium Constant

Reversible Reactions and The Concept of Equilibrium

- Some reactions can go both forward or in reverse. These are called *reversible reactions.*
- A _________________ can be calculated for both the forward reaction and the reverse reaction.
- Chemical equilibrium occurs when a reaction and its reverse reaction proceed at the same _________________. As many molecules are being made as broken.

Chemical equilibrium

- state where concentrations of products and reactants remain _________________
- equilibrium is _________________ (reaction is ________________________)
- any chemical reaction in a closed vessel will reach equilibrium
- at equilibrium, forward reaction rate = reverse reaction rate
- _________________ are equal, _____________________________ are constant



$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The Reaction Quotient, Q

- For the general reversible reaction: $aA + bB \rightleftharpoons cC + dD$, the reaction quotient expression is ALWAYS

  ____:

$$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- Q is the ratio of products to reactants and allows us to examine the ______________________________, whether nearer the products or nearer the reactant.

- Because equilibrium depends on rates, the collision theory tells us that it only includes particles that can actually ______________________.
- For this reason, the reaction quotient expressions only include ______________________or ____, and NOT ______________________or ______________________.
- The ______________________for *Q (and later K)* depend on the reaction. They are usually not used.

Write the Q expression for the following reactions:

1. H_2 (g) + F_2 (g)⇄ 2HF (g)

2. Mg (s) + 2CuCl (aq) ⇄ 2Cu (s) + $MgCl_2$ (aq)

3. $2H_20$ (l) ⇄ $2H_2$ (g) + O_2 (g)

4. Cl_2 (g) + 2Na (s) ⇄ 2NaCl (s)

The Equilibrium Expression (Law of Mass Action)

- For the general reversible reaction: $aA + bB \rightleftharpoons cC + dD$, the reaction quotient expression is ALWAYS ____:

$$K_{eq} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- K_{eq} is the equilibrium _________________________ - the ratio of products to reactants when the system is at EQUILIBRIUM.
- If a system IS at equilibrium, $Q = K_{eq}$. The forward and reverse rates are equal.
- If a system is NOT at equilibrium, $Q \neq K_{eq}$. The reaction will shift until $Q=K_{eq}$.

Gas Systems and Pressures in K

- For gas systems, _________________________ can be used in equilibrium expressions. The equilibrium constant is called _______. Using the same mass action equation as above, for the reaction:

$$aA\ (g) + bB\ (g) \rightleftharpoons cC\ (g)\ dD\ (g)$$

- the K_p expression becomes:

$$K_P = \frac{P_C{}^c P_D{}^d}{P_A{}^a P_B{}^b}$$

- NOTE: P = partial pressure at equilibrium in _________________________.
- Don't use brackets!(you may see this without parentheses)
- Notice that pressure is related to concentration:
- Kc involves concentrations while Kp involves pressures. Be sure to select the correct one when solving a problem. *** you will NOT have to convert from Kc to Kp.
- Write a Kp for sample # 3 from the previous slide.

$$2H_20\ (l) \rightleftarrows 2\ H_2\ (g) + O_2\ (g)$$

K is a CONSTANT.

- This means that for a given reaction at a given _______________________, the ratio of P/R at equilibrium will always be the same. For any reaction, *K* only changes with _______________________ (not with concentration or pressure).

Practice: Write an Equilibrium Expression

$$4\ NH_3(g) + 7\ O_2(g) \leftrightarrows 4\ NO_2(g) + 6\ H_2O(g)$$

If we know equilibrium concentrations, we can calculate the equilibrium constant, K_c

- *K* changes with temperature (not with concentration or pressure).
- A *K* value is only true for one reaction at one specific temperature.
- Change the temperature and *K* will change.

Properties of the Equilibrium Constant

- Consider the rxn: $aA + bB \leftrightarrows cC + dD$: $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

  For the reverse reaction: $cC + dD \leftrightarrows aA + bB$ $K_c' = \frac{[A]^a[B]^b}{[C]^c[D]^d}$

- for any reaction: ______________________

Practice: for the reaction: A $\leftrightarrows$ B, $K_{eq} = 3.4$

Write the expression for $K_{eq.}$

Write the reverse reaction, the corresponding equilibrium expression, and calculate its value.

- If the original reaction is multiplied by some factor to give:

  $naA + nbB \leftrightarrows ncC + ndD$:

  $K_c' = \frac{[C]^{nc}[D]^{nd}}{[A]^{na}[B]^{nb}}$

  for any reaction: ______________________

Practice: for the reaction: A $\leftrightarrows$ B, $K_{eq} = 3.4$

For the reaction 2A $\leftrightarrows$ 2B Write the corresponding equilibrium expression, and calculate its value.

- For a 2-step reaction with K_{c1} and K_{c2} as K_c values for each step, the K_c for the overall reaction is $K_c = K_{c1} \times K_{c2}$.

Practice:

Suppose that for reaction 1: A + B $\rightleftarrows$ C $K_1 = 0.10$

And that reaction 2: C + B $\rightleftarrows$ D $K_2 = 4.5$

What is the overall reaction? What is the value of K for the overall reaction?

Practice: For the reaction: AgCl (aq)+ Na (s) $\rightleftarrows$ NaCl (aq) + Ag (s) , K = 5.7×10^{11},

Write the equilibrium expression

Calculate K for the reaction: 2AgCl (aq)+ 2Na (s) $\rightleftarrows$ 2NaCl (aq) + 2Ag (s)

Calculate K for the reaction: NaCl (aq) + Ag (s) $\rightleftarrows$ AgCl (aq)+ Na (s)

7.5 Magnitude of the Equilibrium Constant

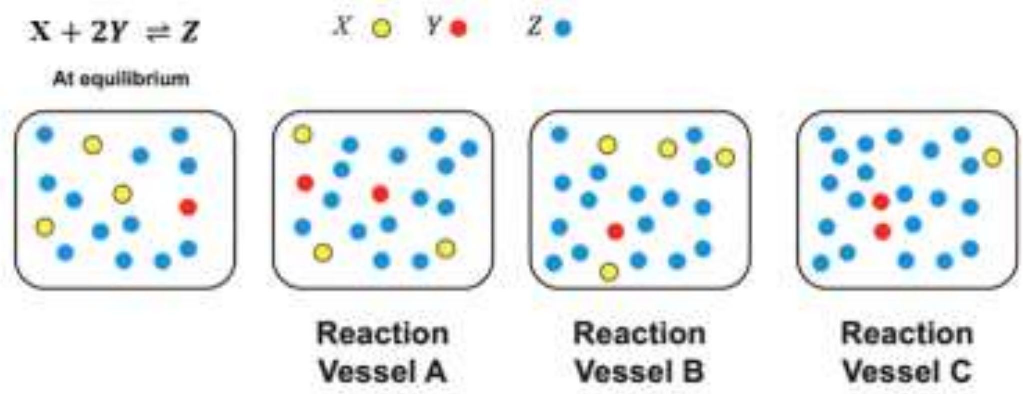
7.7 Calculating Equilibrium Concentrations

7.8 Representations of Equilibrium

Determining if a reaction is at the Equilibrium Position

- Every reversible reaction has an equilibrium constant, K that is temperature specific, in other words, if you change the _____________________, you change _____________________ for that reaction
- At equilibrium, _____________________
- A Q value can be used to determine if a system is at equilibrium, and also what _____________________ (forward or reverse) it will be _____________________to reach equilibrium.
- Q- "_____________________" whether system is at EQ. Use given concentrations to calculate a Q value using the K expression, and compare to the K.

Practice: Which of the following systems are at equilibrium?



Using Q and K

Practice: For the synthesis of ammonia, the value of K is 6×10^{-2} at 500°C. In an experiment, 0.50 mol of $N_2(g)$, 1.0×10^{-2} mol of $H_2(g)$, and 1.0×10^{-4} mol of $NH_3(g)$ are mixed at 500°C in a 1.0 L flask. In which direction will the system proceed to reach equilibrium?

Multiple Choice Practice FTW!

Use the following information to answer #3-5: A sample of H_2S gas is placed in an evacuated, sealed container and heated until the following decomposition reaction occurs at 1000 K:

$$2\,H_2S(g) \leftrightarrow 2\,H_2(g) + S_2(g) \qquad K_c = 86.2$$

3. Which of the following represents the equilibrium constant for this reaction?

a. $K_c = \frac{[H_2]^2[S_2]}{[H_2S]^2}$ c. $K_c = \frac{2[H_2][S_2]}{2[H_2S]}$

b. $K_c = \frac{[H_2S]^2}{[H_2]^2[S_2]}$ d. $K_c = \frac{2[H_2S]}{2[H_2][S_2]}$

4. Initially, $[H_2S]$ = 0.1 M, $[H_2]$ = 0.01 M, and $[S_2]$ = 0.01 M. Which way will the reaction shift and why?
   a. The reaction will shift to make more products because $Q < K$.
   b. The reaction will shift to make more reactants because $Q > K$.
   c. The reaction will shift to make more products because $Q>K$
   d. The reaction will shift to make more reactants because $Q<K$

5. If, at a given point in the reaction, the value for the reaction quotient *Q* is determined to be 2.5×10^8, which of the following is occurring?
   a. The concentration of the reactant is decreasing while the concentration of the products is increasing.
   b. The concentration of the reactant is increasing while the concentration of the products is decreasing.
   c. The system has passed the equilibrium point, and the concentration of all species involved in the reaction will remain constant.
   d. The concentrations of all species involved are changing at the same rate.

6. Under equilibrium conditions, 0.60 moles of A, 0.60 moles of B and 0.60 moles of C are present in a 3.1 L solution for the reaction shown below. Determine the value of the equilibrium constant, *K*.

$$2\,A + B \leftrightarrow 2\,C$$

a. $K = 17$ b. $K = 5.2$ c. $K = 1.7$ d. $K = 0.52$

Consider: moles/liter must be used for aqueous systems, but they can also be used for gases

At a 298 K, a 3.0-L flask contains 2.4 mol of Cl_2 , 1.0 mol NOCl, and 4.5×10^{-3} mol NO. Calculate K_{eq} at this temperature for the following reaction (assuming the reaction is at equilibrium):

$$2\,NOCl\,(g) \rightleftarrows 2\,NO\,(g) + Cl_2\,(g)$$

Write an equilibrium expression for the reaction and solve for the value of K_c at 298 K.

What does this K_c value tell us about the system at equilibrium?

Using K to find concentrations

The reaction: H_2 (g) + F_2 (g)$\rightleftarrows$ 2HF (g) has an equilibrium constant of Kc = 16 at 400°C. If the system at equilibrium was shown to have $[H_2]$ = $[F_2]$ = 1.0 M, What is the [HF]?

At 400°C, does this reaction favor the products or reactants? How do you know?

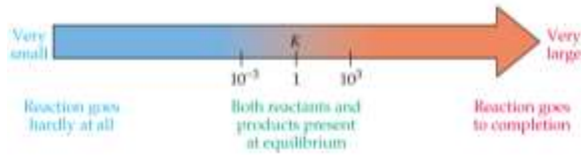
Suppose in another reaction of H_2, F_2, and HF, also at 400°C, we increase the concentration of reactants and when equilibrium was reestablished, the new $[H_2]$= $[F_2]$ = 2.0 M. What would be the [HF] under these circumstances?

Equilibrium position

<u>Equilibrium position</u> - a set of equilibrium concentrations that establish a ________________________________ ____ equal to the equilibrium constant.

- Can be visualized on a number line:

- The ____________________________ of the K value can tell you something about the reaction
- For a reaction with K>1, this means that there are more ____________________________ than ____________________________ at equilibrium. This reaction is said to 'favor the ____________________________ at equilibrium.
- For a reaction with K<1, this means that there are more ____________________________ than ____________________________ at equilibrium. This reaction is said to 'favor the ____________________________ at equilibrium.
- Kinetic Control: The size of K and the time needed to reach equilibrium are not directly related. (example: Rusting or 2Fe + 3 O_2 $\rightleftharpoons$ Fe_2O_3)
- K values of two reactions cannot always be directly compared because stoichiometry differs.



- If the value for K is very ___________________________, reactants are present in great excess at equilibrium and the reaction ____.
- If the value for K is very ___________________________, products are present in great excess at equilibrium, and the reaction is said to ____. (Remember that a large K does not necessarily mean the reaction is fast!)
- Values for K in the range of 0.001 to 1000 describe reactions where reactants and products are ___________________________ in significant quantities at equilibrium.
- The size of K determines how to _________________________ an equilibrium problem!

Practice: Consider the reaction $2NOCl(g) \leftrightarrows 2NO(g) + Cl_2(g)$ at 35ºC. Various amounts of $NOCl(g)$, $NO(g)$, and $Cl_2(g)$ are mixed in a 10.0 L flask. After the system has reached equilibrium the concentrations are observed to be:

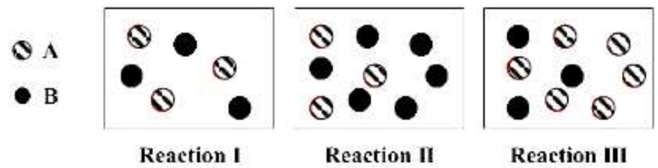
$[Cl_2] = 1.52 \times 10^{-1}\ M$ $[NO] = 4.00 \times 10^{-3}\ M$ $[NOCl] = 3.96 \times 10^{-1}\ M$

a. Calculate the value of K for this system at 35ºC.

b. What can we conclude about the reaction progress from the value of K?

Let's Practice!

1. The following boxes represent reactions of A ⇋ B at equilibrium.

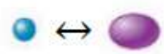


For which reaction shown above is K smallest?

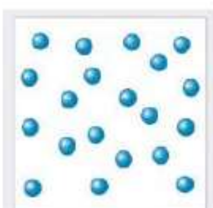
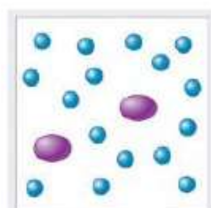
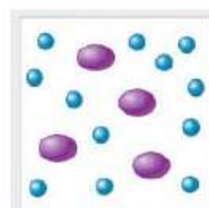
   a. Reaction I
   b. Reaction II
   c. Reaction III
   d. Cannot be determined.

Consider the following reversible reaction to answer #2–3.

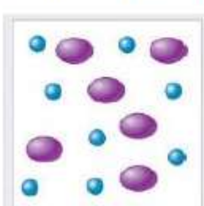
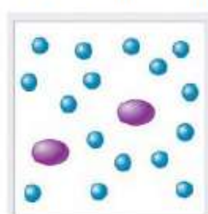
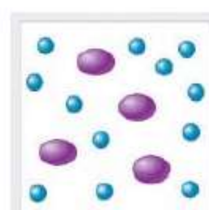
$$A \leftrightarrow B$$



2. Which of the following equilibrium systems has the largest value of K?

a.  b.  c. 

3. Which of the following equilibrium systems has the smallest value of K?

b.  b.  c. 

7.9 Intro to LeChatlier's Principle

7.10 LeChatlier and the Reaction Quotient, Q

Le Chatlier's Principle

- When a system is at equilibrium, ________________________.
- If a stress is applied to a system at equilibrium making ________________________, the system will ________________________ in the direction that relieves the stress and returns ________________________.
- A stress is any kind of change in a system ________________________ that upsets the equilibrium (makes K ≠ Q).

- 3 types of stress
  - Changes in concentration
  - Changes in Pressure-Volume
  - Changes in Temperature

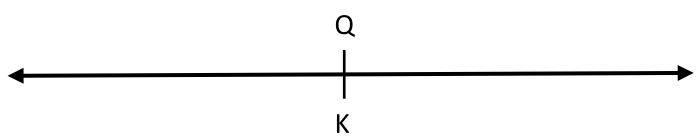
STRESS: Changes in concentration

Consider the reaction at equilibrium: $A\ (g) + B\ (g) \rightleftharpoons C(g)$

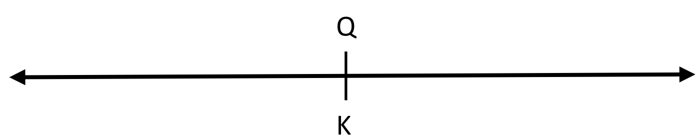
Write an equilibrium expression:

- At equilibrium, Q=K (1)
- An increase in concentration of a reactant will cause a *increase/decrease* in Q (2), disturbing the equilibrium of the system. The system will shift *forward/reverse* to restore equilibrium (3). As a result, the concentration of the reactants *increase/decrease* and the concentration of the products *increase/decrease.*

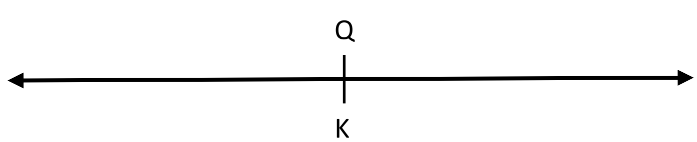
Let's put it on a number line to represent the steps:



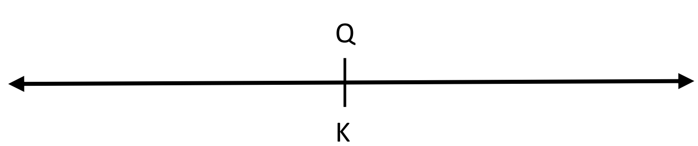
- An increase in concentration of any product will cause Q to increase/decrease equilibrium to shift *forward/reverse* to *increase/decrease* the concentration of reactants and *increase/decrease* the concentration of the products.



- A decrease in concentration (removal) of a reactant will cause Q to increase/decrease, causint the reaction to shift forward/reverse to restore equilibrium. . The system will shift forward/reverse to restore equilibrium. As a result, the concentration of the reactants increase/decrease and the concentration of the products increase/decrease.



- An decrease in concentration of any product will cause Q to increase/decrease equilibrium to shift forward/reverse to increase/decrease the concentration of reactants and increase/decrease the concentration of the products.



OVERALL: A change in concentration of reactant or product will affect the value of Q, and not affect the value of *K*.

Practice: Consider the reaction: $N_2(g) + 3H_2(g) \leftrightarrows 2NH_3(g)$

If the system is at equilibrium, what direction of shift will each action cause?

a. addition of N_2

b. addition of NH_3

c. addition of H_2

d. removal of NH_3

STRESS: Changes in pressure or volume- only gasses affected

- Changes in pressure only affect equilibrium systems having _________________________ products and/or reactants.
- Increasing the pressure/decreasing the volume of a gaseous system will cause equilibrium to shift to the side with _________________________ gas particles.
- Decreasing the pressure/increasing the volume of a gaseous system will cause equilibrium to shift to the side with _________________________ gas particles.
- If the system has the _________________________ number of moles of gas on each side, changes in pressure do not affect equilibrium.
- Adding an _________________________ gas does not affect equilibrium since the partial pressures of the gases in the reaction are not affected.
- Changing pressure does not affect the _________________________ of the equilibrium constant.

Example: determine the direction of shift in response to each action

$P_4(s) + 6Cl_2(g) \leftrightarrows 4PCl_3(l)$

a. increase container volume

b. decrease container volume

c. add argon gas

$PCl_3(g) + Cl_2(g) \leftrightarrows PCl_5(g)$

a. decrease container volume

b. add helium gas

$PCl_3(g) + 3NH_3(g) \leftrightarrows P(NH_2)_3(g) + 3HCl(g)$

a. increase container volume

STRESS: Changes in temperature

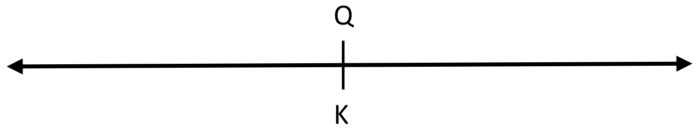
- Changes in temperature may easily be treated as changes in concentration if you think of _________________________ as a product (exothermic rxn) or a reactant (endothermic rxn).

Example: $B_2(g) + C_2(g) \rightleftarrows 2BC(g)$ $\Delta H = +21 kJ/mol$

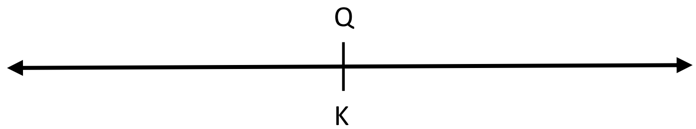
Is heat a reactant or product? Place it on the correct side.

Exothermic reaction: R ⇄ P + Δ

- An increase in temperature of an exothermic reaction will cause equilibrium to shift *forward/reverse*. K will *increase/decrease.*



- A decrease in temperature of an exothermic reaction will cause equilibrium to shift *forward/reverse*. K will *increase/decrease.*



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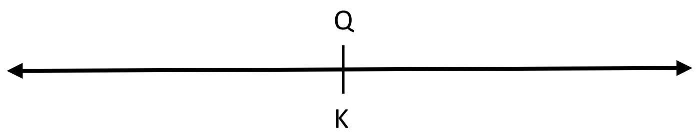

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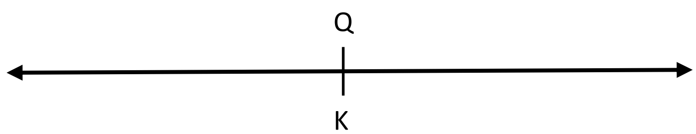

- For the reaction: $B_2(g) + C_2(g) \rightleftarrows 2BC\ (g)$ $\Delta H = +21kJ/mol$, if the temperature is increased, the reaction will shift ____________________ and K will _________________

Example: $N_2(g) + 3H_2(g) \rightleftarrows 2NH_3(g)$ $\Delta H = -21^oC$

- Is heat a reactant or product?
- What happens if T increases? To the concentration of reactants? To concentration of products? To K?

- What happens if T decreases? To the concentration of reactants? To concentration of products? To K?

Catalysts & equilibrium

- With a catalyst, reactions reaches equilibrium quickly, but _______________________ in the amount of product or the value of K.
- In other words, catalyst make you reach equilibrium faster, but doesn't __________________ the system

72. Coal can be used to generate hydrogen gas (a potential fuel) by the endothermic reaction:

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

If this reaction mixture is at equilibrium, predict whether each disturbance will result in the formation of additional hydrogen gas, the formation of less hydrogen gas, or have no effect on the quantity of hydrogen gas.

a. adding more C to the reaction mixture
b. adding more H_2O to the reaction mixture
c. raising the temperature of the reaction mixture
d. increasing the volume of the reaction mixture
e. adding a catalyst to the reaction mixture
f. adding an inert gas to the reaction mixture

- Which ONE affects K? How is K affected- increase or decrease?

How to Answer FR Questions about Le Châtelier

1. Identify stress (change) to the system.
2. Identify effect of stress on the equilibrium system: shift right, shift left, or no shift?
3. Explain how the shift you identified will counteract the original stress to re-establish equilibrium.
4. Connect your explanation to the question asked (aka, answer the question. ;D)

Example #1: If the temperature of an exothermic reaction changes from 25oC to 50oC, what is the effect on equilibrium constant, K?

Increasing the temperature of an exothermic reaction will cause the system to shift left to use up added heat, increasing [reactants] and decreasing [products] to re-establish equilibrium, decreasing K.

Example #2: For the reaction A(g) + B(g) ⇌ C(g), adding more A(g) to the reaction vessel will have what effect on the concentration of B(g)?

Adding more A(g) will increase [A], causing the system to shift right to use up some of the extra A and re-establish equilibrium. Using up A will also use up some B(g), so [B] will decrease.

Example #3: For the reaction A(g) + B(g) ⇌ C(g), *removing some B(g) from the reaction container will have what effect on the reaction mixture?*

Removing some B(g) will decrease the concentration of a reactant, causing the system to shift left to make more reactants and re-establish equilibrium.

Free Response Practice!

1. In the following reaction at chemical equilibrium, what is the effect on the reaction mixture if the volume of the reaction container is decreased? Explain. $N_2(g) + O_2(g) \rightleftharpoons 2\ NO(g)\ \Delta H° = 181\ kJ$

2. Consider the following endothermic reaction at chemical equilibrium. When the temperature of the reaction mixture is increased, what is the effect on CaCO3(s)? Explain. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

3. Adding additional KCl will have what effect on the reaction mixture, if the mixture was originally at chemical equilibrium? Explain. $2\ KClO_3(s) \rightleftharpoons 2\ KCl(s) + 3\ O_2(g)$

Review:

For the reaction: $2\ NO_{2\ (g)} \rightleftharpoons N_2O_{4\ (g)}$, ΔH_{rxn} = -177 kJ/mol, and K_p = 0.785 at 387 K.

a) Write an expression for K_p.

b) If a previously evacuated reaction vessel was filled with 0.50 atm of NO_2 and 0.50 atm of N_2O_4, is the system at equilibrium? If not, which way would the reaction shift to reach equilibrium?

c) NO_2 is brown while N_2O_4 is colorless. If the reaction is light brown at equilibrium, predict the direction of shift, and changes to the color of the reaction.

   i. The reaction is heated

   ii. The gases are placed in the container half the size.

   iii. More $N_2O_{4\ (g)}$ is added

   iv. Will the value of K increase or decrease when this reaction is cooled?

7.7 Calculating Equilibrium Concentrations

Equilibrium and RICE Tables

The stoichiometry of a chemical reaction and the associated K value, can be used to determine the concentrations of products or reactants for a reaction at equilibrium based on the initial values for the reacting species.

We can use a RICE Table to keep track of our reaction.

- R- Reaction
- I- Initial
- C- Change
- E- Equilibrium

What's new about these RICE table:

- E = Equilibrium Concentration ~*OR*~ Pressure in atm
- There might be some of ALL the reactants and products left
- Quantities are in mol/L (M) for Kc or atm for Kp, NOT moles

Given some initial and equilibrium data for a reaction, the value of K can be calculated.

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Practice: Consider the following reaction:

$$CO(g) + 2\ H_2\ (g) \rightleftharpoons CH_3OH\ (g)$$

A reaction mixture at 780 ° C initially contains [CO] = 0.500 M and [H_2] = 1.00 M. At equilibrium, the CO concentration is found to be 0.15 M. Calculate the concentrations of all species at equilibrium. What is the value of the equilibrium constant? Does this reaction favor the reactants or the products?

<u>EXAMPLE:</u> A rigid vessel at a temperature of 25ºC initially has a partial pressure of NO equal to 0.526 atm and a partial pressure of Br_2 equal to 0.329 atm. At equilibrium the partial pressure of Br_2 is 0.203 atm.

(a) Calculate K_p and K_c for the reaction: $2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$

(b) Calculate the total pressure in the container once equilibrium has been established.

Using K to find Concentrations of Reactants and Products

Size of K and Reaction Position:

- If a reaction has a very LARGE K value ($K > 10^5$), the reaction is considered to leave negligible reactants at equilibrium, or the reaction goes TO COMPLETION.
- If a reaction has a very small K value ($K < 10^{-5}$), the reaction is considered to produce very few products at equilibrium. The tiny product concentrations can be calculated using an approximation for the amount of reaction.

Example: K is LARGE (>10^5)

Rxn goes to completion…. Solve as a normal STOICH problem! Remember to consider limiting reactants.

Practice: Consider the reaction: 2 NO (g) + O_2(g) ⇌ 2 NO_2

If $K = 1.0 \times 10^{23}$ at 500°C, calculate the molarity of NO_2 produced when 1.0 M of NO reacts with of 0.25 M O_2.

Small K values: The case of the vanishing x: useful when K is small

When we know that a reaction MUST proceed to the products, even in tiny amounts, we will use "x" to represent the change. Using the mole- mole ratio, then this might be change and be 2x or 3x etc.

We can then complete the equilibrium values in terms of "x", and also apply an approximation for when x is very small (<5% of the initial value).

Practice: Finding Equilibrium Concentrations from Initial Concentrations and the Equilibrium Constant
Consider the reaction: N_2 (g) + O_2(g) ⇌ 2 NO $Kc = 2.0 \times 10^{-5}$ at 2000ºC
A reaction mixture initially contains [N_2] = 1.800 M and [O_2] = 1.500 M. Find the equilibrium concentrations of the reactants and product at this temperature.

The Vanishing x - Checking the Approximation and Refining as Necessary

- When solving an equilibrium problem, some +*x* and -*x* values can be treated as negligible if it is less that 5% of the number that it was to be subtracted from or added to. If *x* is not negligible, the quadratic equation must be used. (Never the case on an AP test)
- We can check our approximation by comparing the approximate value of *x* to the initial concentration. If the approximate value of *x* is less than 5% of the initial concentration, the approximation is valid.
- If $\frac{approximate\ x}{initial\ concentration}$ x 100% , 5%, the approximation is valid

Practice: Consider the reaction for the decomposition of dihydrogen sulfide

$$2\ H_2S \rightleftharpoons 2\ H_2 + S_2 \qquad K_c = 1.67 \times 10^{-7} \text{ at } 800°C$$

A 0.500 L reaction vessel initially contains 0.125 mol of H_2S at 800 °C. Find the concentrations of H_2 and S_2 at equilibrium.

What if x isn't negligible??
The Perfect Square: if keeping "−x" is necessary, see if the problem is a perfect _____________ and thus easy to solve using algebra.
Example: Consider the reaction: H2(g) + I2(g) ⇌ 2 HI(g), Kp = 64 (at 25oC). A reaction mixture at 25°C initially contains $PH_2 = PI_2 = 0.20$ atm. Find the equilibrium partial pressures of H_2, I_2, and HI.

Multiple Choice Practice!

1. Here is a general reaction with a K value of 16: $A(aq) + B(aq) \rightleftharpoons 2\ C(aq)$.
   Initially, [A] = [B] = 2.0 M. Solve for the equilibrium concentration of each substance.

   a. [A] = [B] = 0.67 M, [C] = 1.3 M c. [A] = [B] = 0.67 M, [C] = 2.7 M

   b. [A] = [B] = 1.6 M, [C] = 0.88 M d. [A] = [B] = 0.50 M, [C] = 3.0 M

2. Consider the following equilibrium: $Cl_2(g) + 2\ NO(g) \rightleftharpoons 2\ NOCl(g)$. Initially, the reaction was started by adding 10.0 *M* NOCl gas to a reaction vessel. At equilibrium, [NO] = 2.0 *M*. What is the value of K_c?

   a. 0.63 b. 2.3 c. 4.0 d. 16

3. The reaction below came to equilibrium at a temperature of 100°C. At equilibrium the partial pressure due to NOBr was 4 atm, the partial pressure due to NO was 4 atm, and the partial pressure due to Br_2 was 2 atm. What is the equilibrium constant, K_p, for this reaction at 100°C?

   $$2\ NOBr(g) \rightleftharpoons 2\ NO(g) + Br_2(g)$$

   a. ¼ b. ½ c. 1 d. 2

4. Consider the following: $2\ SO_2\ (g) + O_2(g) \rightleftharpoons 2\ SO_3\ (g)$. Initially, 0.030 mol SO_2 and 0.030 mol O_2 are placed into a 1.0 L container. At equilibrium, there is 0.020 mol O_2 present. What is the [SO_2] at equilibrium?

   a. 0.010 mol/L b. 0.020 mol/L c. 0.030 mol/L d. 0.040 mol/L

Use the information below to answer #5-7.

Reaction 1: $NOBr\ (g) \rightleftharpoons NO\ (g) + ½\ Br_2\ (g)$ $K_c = 3.4 \times 10^{-2}$
Reaction 2: $2\ NOCl\ (g) \rightleftharpoons 2\ NO\ (g) + Cl_2\ (g)$ $K_c = 1.6 \times 10^{-5}$
Reaction 3: $2\ NO\ (g) + 2\ H_2\ (g) \rightleftharpoons N_2\ (g) + 2\ H_2O\ (g)$ $K_c = 4.0 \times 10^{6}$
Reaction 4: $N_2\ (g) + O_2\ (g) \rightleftharpoons 2\ NO\ (g)$ $K_c = 4.2 \times 10^{2}$

5. For a reaction involving nitrogen monoxide inside a sealed flask, the value for the reaction quotient (Q) was found to be 1.1×10^{2} at a given point. If, after this point, the amount of NO gas in the flask increased, which reaction is most likely taking place in the flask?

   a. reaction 1 b. reaction 2 c. reaction 3 d. reaction 4

6. Which of these reactions proceeds at the slowest rate?

   a. reaction 1 b. reaction 2 c. reaction 3 d. cannot be determined

7. For reaction #3, equimolar amounts of N_2 gas and H_2O gas are allowed to come to equilibrium in a sealed reaction vessel. Which of the following must be true at equilibrium?

   I. [N_2] must be less than [H_2O].
   II. [N_2] must be greater than [H_2O].
   III. [NO] must be greater than [H_2].

   a. I only b. II only c. I and III d. II and III